

CONTACT

LOCATION
Odesa, Ukraine · no relocation

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SKILLS

Frontend
TypeScript · React · Next.js · Vue 3 · Angular · Redux · Three.js · Tailwind

Backend
Node.js · Go · NestJS · Express · Hono · Django · Flask · Rails · Spring Boot · ASP.NET Core · REST · GraphQL · gRPC

Real-time
WebRTC · Socket.IO · WebSocket · Yjs CRDT · MQTT

Data & AI
Python · NumPy · Pandas · scikit-learn · XGBoost · TensorFlow · PyTorch · Matplotlib

Databases
PostgreSQL · MySQL · MongoDB · SQLite · Redis · Firebase · Drizzle

Security
Pentesting · OWASP Top 10 · WAF · Traffic analysis · Burp Suite · Nmap · Wireshark · Snyk · Semgrep · SonarQube

Tools & Games
Docker · Git · GitHub Actions · Vite · Jest · Unity / C#

Oleksii Silveistruk

Full-Stack Engineer · Applied ML & Cybersecurity Researcher

Full-stack developer who builds complete products end to end — from 3D graphics and real-time systems to ML-based security. Published researcher: a journal article on chess-engine search optimization and a conference paper on an ML Web Application Firewall (99.76% accuracy, 0.34% FPR). Master's student in cybersecurity with 78 public repositories on GitHub.

78	2	120×	99.76%
GITHUB REPOS	2026 PUBLICATIONS	FEWER NODES SEARCHED	WAF ACCURACY

RESEARCH & PUBLICATIONS

JOURNAL ARTICLE · 2026

Optimization of the Minimax Algorithm with Alpha-Beta Pruning for Chess Artificial Intelligence Systems

O. R. Silveistruk · *Informatics. Culture. Technology*, Vol. 3, No. 1 (3), pp. 179–188 · Sep 2026 · in Ukrainian · DOI 10.15276/ict.03.2026.15 · Article page · PDF

Python minimax engine extended with iterative deepening, Zobrist-hashed transposition tables, MVV-LVA move ordering, killer/history heuristics and quiescence search. Evaluated nodes cut ~120× at depth 5; decision time 92.4 s → 0.43 s; up to 96.5% wins at level 0 against a reference engine.

CONFERENCE PAPER · AUG 2026

A System for Filtering HTTP Traffic of Web Applications (WAF) Using Machine Learning

O. R. Silveistruk · *Perspective Directions of Information Protection*, VIII Int. Sci.-Pract. Conf., Odesa, 26–28 Aug 2026, pp. 106–109 · in Ukrainian · Proceedings (PDF) · Certificate

Intelligent reverse-proxy WAF: 51-feature request vectors (incl. Shannon entropy), SMOTE balancing, XGBoost / Random Forest / SVM / MLP ensemble with hybrid signature cross-validation. On 12,000 fuzzed transactions: **99.76% accuracy, 0.34% false positives, ≈15 ms latency** (p99 14.8 ms at 1,500 RPS).

Earlier conference papers: Current Trends in Scientific Research Development, Boston, Dec 2024 (pp. 269–274) · European Congress of Scientific Achievements, Barcelona, Dec 2024 (pp. 206–210) · 79th SUITT Sci.-Tech. Conf., Nov 2024 (pp. 185–187) · 80th SUITT Sci.-Tech. Conf., Feb 2026 (pp. 296–300).

SELECTED PROJECTS

Online Compiler · own execution engine — 2026

Browser IDE without Judge0/Piston: 41 languages, 615 reference snippets auto-checked against expected output, real-PTY interactive console, debugger for 7 languages, Scratch 3 / Blockly, Arduino Uno emulation in-browser and ESP32 in QEMU. CI + nightly runs.

Django · PostgreSQL · Docker · Monaco · xterm.js · WebSocket

Chess 3D AI · basis of the journal article — 2026

Four-package monorepo: custom TypeScript engine (alpha-beta, quiescence, opening book, 5 levels, Web Worker), procedural Three.js board with 3 themes and GSAP effects, Express + SQLite backend with Elo and analytics, Vue 3 client.

TypeScript · Three.js · Vue 3 · Express · SQLite

ZTNA-ml · zero-trust access with ML trust scoring — 2026

WireGuard zero-trust panel that scores every node's trust from its traffic: rules + LSTM autoencoder (ONNX, ~0.2 ms/window) catch brute force, scanners and exfiltration, and the gateway auto-quarantines nodes via nftables. Recall 0.863, ROC-AUC 0.974, FPR 0.51%.

Python · ONNX Runtime · Django REST · Next.js · nftables · Redis Streams · Docker

EDUCATION

M.Sc. Cybersecurity & Information Protection
State University of Intelligent Technologies and Communications
Sep 2025 – Dec 2026 (expected)

B.Sc. Computer Science
State University of Intelligent Technologies and Communications
Graduated 2025 · score 100
↗ Diploma

LANGUAGES

English
B1 · Intermediate

CERTIFICATIONS

EPAM
Front-End Fundamentals course certificate
↗ Verify certificate

Odesa, Aug 2026
VIII Int. Sci.-Pract. Conf. “Perspective Directions of Information Protection”
↗ Certificate ↗ Proceedings

Boston, Dec 2024
Participant, V Int. Sci.-Pract. Conf. “Current Trends in Scientific Research Development”
↗ Certificate ↗ Conference

Barcelona, Dec 2024
Participant, XII Int. Sci.-Pract. Conf. “European Congress of Scientific Achievements”
↗ Certificate ↗ Conference

ACTIVITIES

Science Skills Booster SUITT 2024
Key team member — “Ukrainian YouTube”
↗ Event report

Innovation Hub hackathons
Built and presented projects at the university
↗ Event report

Freelance
E-commerce websites for multiple clients

FOCUS AREAS

ML security · Game-tree search · Real-time systems · Developer tools · 3D web · IoT

SELECTED PROJECTS (continued)

Mirage · honeynet lab — 2026
Isolated VirtualBox network of three decoy servers (Cowrie, OpenCanary, Dionaea), honeytokens, IDS, Kibana dashboards and an event correlator that maps attack campaigns to MITRE ATT&CK, with a full Kali attack write-up.
Vagrant · Ansible · Docker · Kibana · Python

Go Sales CRM · Go + Vue 3 — 2026
Orders with transactional stock reservation and price snapshots, status workflow with history, customer segments, review moderation, analytics dashboard; UA/EN, responsive, Playwright tests.
Go · chi · pgx · PostgreSQL 17 · Vue 3 · Pinia · ECharts · Docker

CollabBoard · Miro-style whiteboard — 2026
12 canvas tools, draw.io-style connectors, Yjs CRDT real-time collaboration, rich-text documents, workspaces, gRPC gateway.
Next.js 16 · React 19 · Konva · Yjs · gRPC · Drizzle

SocialNet · social network + Mini Social — 2026
JWT + Google OAuth, feed, friends, WebSocket messenger, WebRTC calls, channels. Mini Social: Vue 3 + Express 5 + PostgreSQL monorepo with tests, CI and Docker.
Next.js · Hono · Vue 3 · Express 5 · WebRTC · Drizzle

Monitoring ESP32 · IoT + ML — 2026
MQTT ingestion from ESP32 sensors, live dashboard, XGBoost anomaly detection and forecasting.
Spring Boot 3.2 · React 19 · MQTT · XGBoost · MySQL

Peremoga · aid platform for Ukraine — 2026
Four roles, aid tickets, fundraisers, live updates via ActionCable, ActiveAdmin, Swagger API, UA/EN.
Rails 8.1 · Angular 22 · PostgreSQL 17 · JWT

Unity games · Hogwarts Mini, Soul Knight Clone, Lullaby of Echoes — 2026
Third-person action RPG (36 story stages, 15 spells, 14 enemy types, 22 castle zones); a procedural roguelike (49 weapons, 60+ enemies, 6 bosses) and a narrative RPG — all in Unity 6.
Unity 6 · C#

Also on GitHub: Music Platform (NestJS, Next.js) · LangLearn (ASP.NET Core, Angular) · Mini Excel (TypeScript) · Video Chat (WebRTC) · EduLink (shortener + LMS) · КупиДевайс (PERN) · Shop (Hono, Redis) · North bank (Vue) · Dela notes PWA · GitHub Explorer · Corporate Chats (Flask) · UVP (Django) · Checkers AI.

EXPERIENCE

Junior Frontend Developer · Asferro — Jan – Feb 2022
Migrated an admin dashboard from GraphQL Apollo to REST; worked in a full-stack team until the company closed because of the war.

Production Intern · ISyb — Feb – Mar 2024
Built a full-stack auto-parts e-commerce site with React, Node.js and TypeScript.

Front-End Course · EPAM Systems ↗ certificate — Feb – Mar 2025
Completed the Front-End development course and earned a certificate.

“Ukrainian YouTube” · Science Skills Booster SUITT 2024 — 2024
Student content-sharing platform: React/Next.js (SSR/SSG), NestJS, PostgreSQL + TypeORM, JWT, video upload, categories, likes, comments, subscriptions.